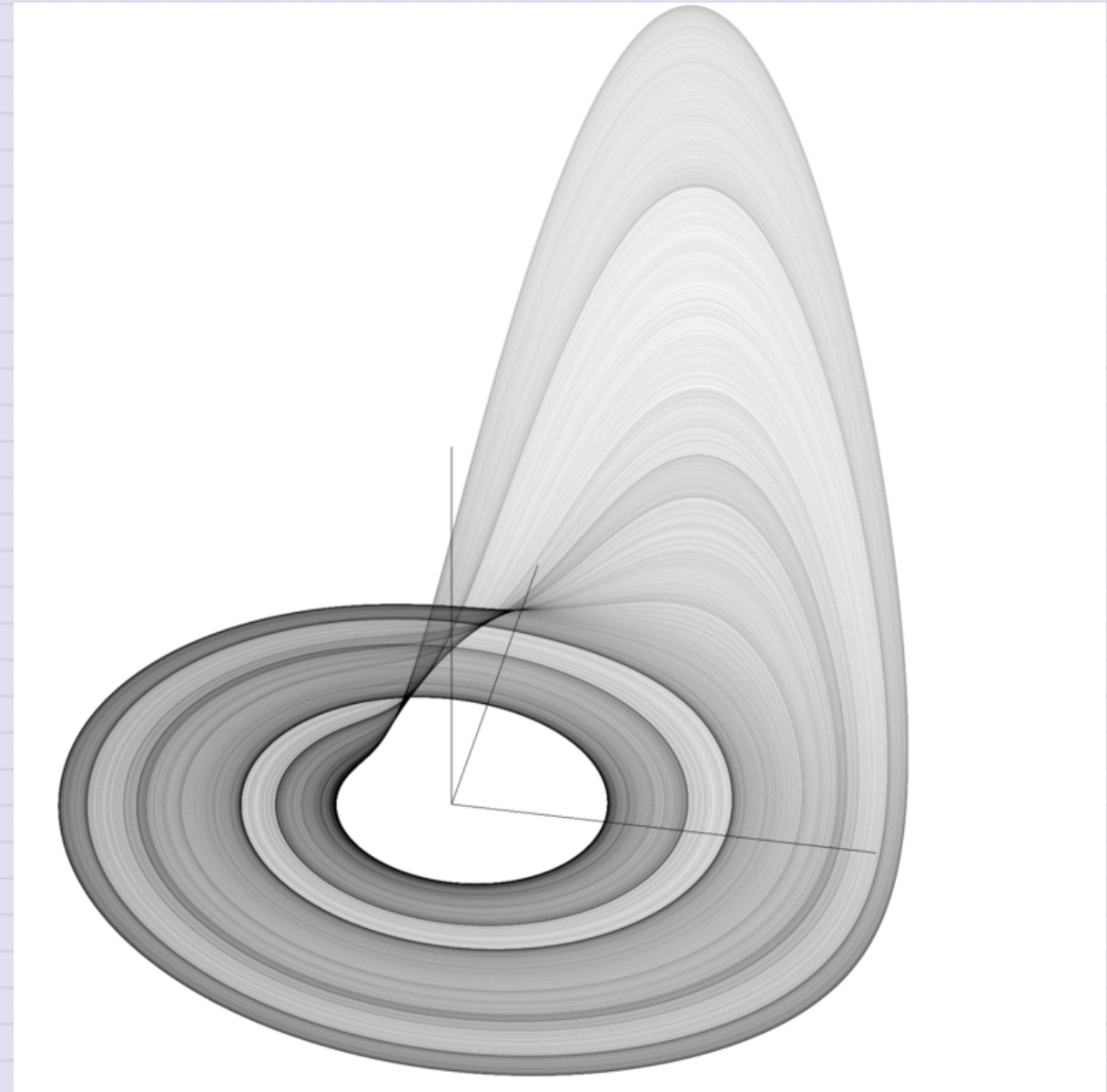


Day 18 - Introduction to Nonlinear Dynamics

Rössler Attractor

$$\begin{aligned}\dot{x} &= -y - z \\ \dot{y} &= x + ay \\ \dot{z} &= b + z(x - c)\end{aligned}$$

**There are no crossings of
the path shown in this
picture**



Announcements

- Midterm 1 is due Oct 10th

Seminars this week

MONDAY, October 6, 2025

Condensed Matter Seminar 4:10 pm, 1400 BPS, In Person and Zoom, Host ~ Tyler Cocker

Speaker: Edoardo Baldini, University Texas - Austin

Title: Nonlinear Coupled Magnonics

Zoom Link: <https://msu.zoom.us/j/93613644939>

Meeting ID: 936 1364 4939

Password: CMP

Seminars this week

WEDNESDAY, October 8, 2025

FRIB Nuclear Science Seminar, 11:00am., FRIB room 1300 and online via Zoom.

Speaker: Dr. Yuhu Zhai of Princeton Plasma Physics Laboratory

Title: Innovation in superconducting magnet technologies: R&D Gaps and opportunities to mature HTS for fusion and other applications

Join Zoom: [https://msu.zoom.us/j/96657965451?](https://msu.zoom.us/j/96657965451?pwd=lsaf23sK5agzaao0Kwaei7AaWHkc4W.1)

[pwd=lsaf23sK5agzaao0Kwaei7AaWHkc4W.1](https://msu.zoom.us/j/96657965451?pwd=lsaf23sK5agzaao0Kwaei7AaWHkc4W.1)

Meeting ID: 966 5796 5451

Passcode: 479842

Seminars this week

WEDNESDAY, October 8, 2025

Astronomy Seminar, 1:30 pm, 1400 BPS, In Person and Zoom, Host~

Speaker: Kyle Kremer, UC San Diego

Title: Globular Clusters: Astronomical factories of gravitational-wave and electromagnetic transients

Zoom Link: [https://msu.zoom.us/j/94228252584?](https://msu.zoom.us/j/94228252584?pwd=zNJdjCNAabNkppOcg5FlAv01ihzLwl.1)

[pwd=zNJdjCNAabNkppOcg5FlAv01ihzLwl.1](https://msu.zoom.us/j/94228252584?pwd=zNJdjCNAabNkppOcg5FlAv01ihzLwl.1)

Meeting ID: 942 2825 2584

Passcode: 541987

Seminars this week

THURSDAY, October 9, 2025

Colloquium, 3:30 pm, 1415 BPS, in person and zoom. Host ~ Xing Wu

Refreshments and social half-hour in BPS 1400 starting at 3 pm

Speaker: Benjamin Jones, University of Texas at Arlington

Title: Single Barium Ion Identification Technologies for Background-Free Neutrinoless DoubleBeta Decay searches -

Zoom Link: <https://msu.zoom.us/j/94951062663>

Password: 2002 Or complete link: <https://msu.zoom.us/j/94951062663?pwd=c48uM25P9UsRVuR74rkOioOWgpoxgC.1>

Seminars this week

FRIDAY, October 10, 2025

QuIC Seminar, 12:30pm, -1:30pm, 1300 BPS, Zoom only

Speaker: Daniel Nino, Xanadu

Title: Quantum Computational Chemistry (With PennyLane)

Full Schedule is at: <https://sites.google.com/msu.edu/quic-seminar/>

For more information, reach out to Ryan LaRose

Seminars this week

FRIDAY, October 10, 2025

FRIB IReNA Online Seminar, 2:00pm., Eastern Time.

Hosted By: Hosted by: Borbala Cseh (Konkoly Observatory)

Speaker: Artemis Spyrou, MSU

Title: Neutron-capture reaction constraints for astrophysical processes Please see website for full abstract.

Please click the link below to join the webinar:

Join Zoom: <https://msu.zoom.us/j/827950260>

Meeting ID:

Passcode: JINA

Reminders: Conservative Forces

- The curl of a conservative force is zero

$$\nabla \times \vec{F}_{cons} = 0$$

- Work done by a conservative force is path-independent

$$\underbrace{\int_{\vec{r}_1}^{\vec{r}_2} \vec{F}_{cons} \cdot d\vec{r}}_{\text{path a}} = \underbrace{\int_{\vec{r}_1}^{\vec{r}_2} \vec{F}_{cons} \cdot d\vec{r}}_{\text{path b}}$$

- Work done by a conservative force around a closed path is zero

$$\oint \vec{F}_{cons} \cdot d\vec{r} = 0$$

Reminders: Conservative Forces

- The work done by a conservative force is equal to the negative of the change in potential energy

$$W = -\Delta U$$

- A conservative force can be expressed as the gradient of a scalar function

$$\vec{F}_{cons} = -\nabla U$$

Reminders: Equilibrium (aka Critical or Fixed) Points

We found equilibrium points by setting the derivative of the potential energy to zero:

$$\frac{dU}{dx} = 0$$

We then determined if these points were stable or unstable by looking at the second derivative of the potential energy:

$$\frac{d^2U}{dx^2} > 0 \quad \text{stable}$$

$$\frac{d^2U}{dx^2} < 0 \quad \text{unstable}$$

Relationship to Differential Equations

By setting $\frac{dU}{dx} = 0$, we are finding the equilibrium points where the force is zero,

$$-\frac{dU}{dx} = F_x = 0$$

If we consider the typical form of a differential equation,

$$m\ddot{x} = F(x)$$

We can see that we are seeking the points where the differential equation is zero,

$$m\ddot{x} = 0$$

This approach is a powerful way to understand the behavior of a system. And we can do so geometrically!

Clicker Question 18-1

Let $\dot{x} = \sin x$. Set up the integral that could be used to solve for $t(x)$.

1. $\int \frac{dx}{\sin x}$

2. $\int \frac{dx}{\cos x}$

3. $\int \frac{dx}{\tan x}$

4. $\int \frac{dx}{\cot x}$

5. ???

Clicker Question 18-2

We can integrate this with $x(0) = x_0$ to find $t(x)$:

$$t(x) = \ln(|\csc x - \cot x|) - \ln(|\csc x_0 + \cot x_0|)$$

Find $x(t)$? 🙄🙄🙄

Instead find the equilibrium points (x^*) of the system. n is an integer.

1. $x^* = 0$
2. $x^* = 0, \pm\pi$
3. $x^* = \pm\pi/2$
4. $x^* = n\frac{\pi}{2}$
5. $x^* = n\pi$

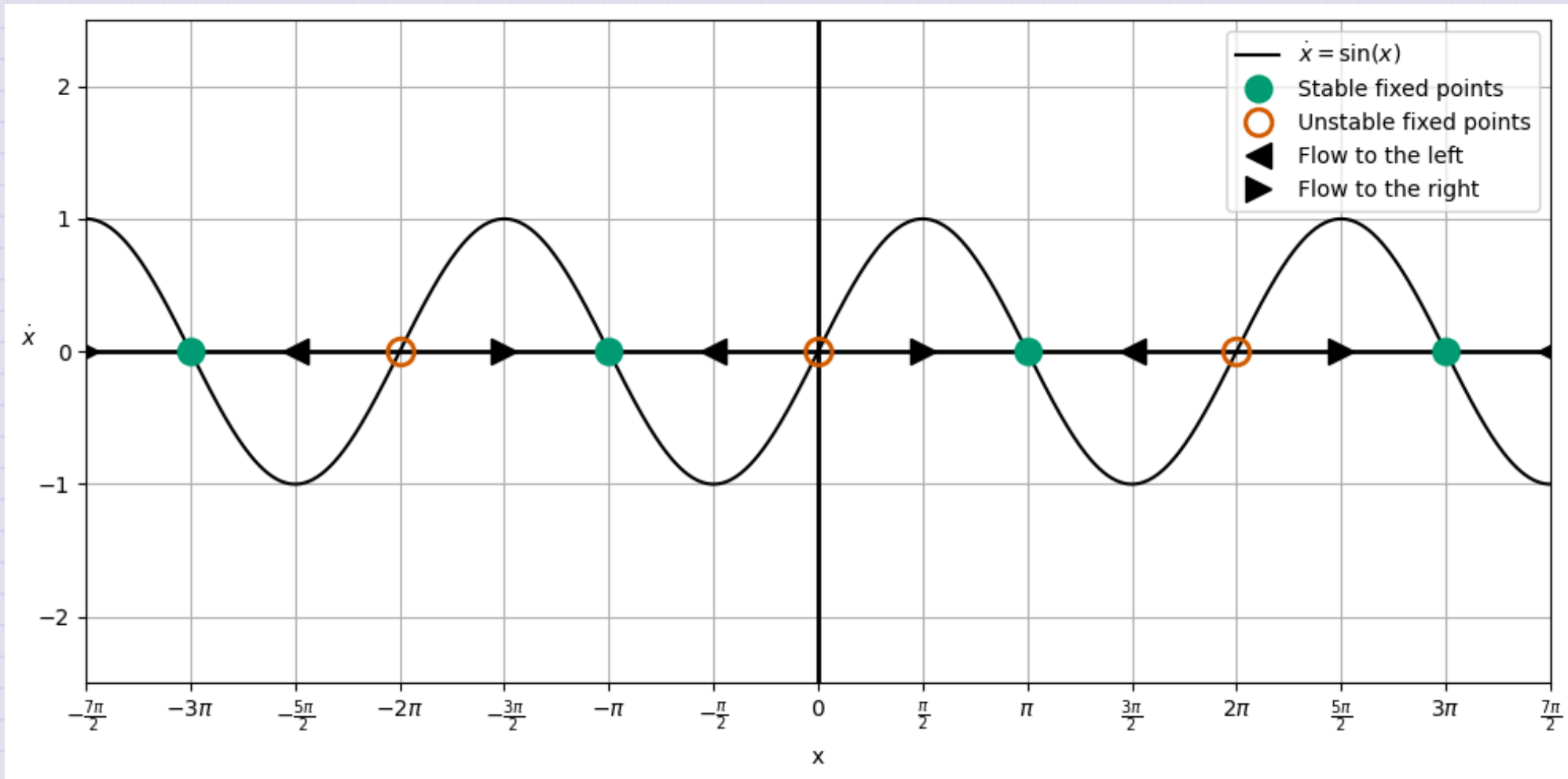
Clicker Question 18-3

Sketch the differential equation $\dot{x} = \sin x$ in the phase space x vs. $\dot{x}$.

1. Note where the plot crosses the x -axis. These are the critical/equilibrium points, x^* . Identify the critical points.
2. By definition, $\dot{x} > 0$ is a "flow to the right" and $\dot{x} < 0$ is a "flow to the left". Sketch the direction of the flow - this should only appear in the x -axis.
3. Look at the flow directions and the critical points. What can you say about the stability of the critical points? We use closed circles for stable points and open circles for unstable points. Add these to your plot.

Click when you and your table are done.

Phase Space Diagram for $\dot{x} = \sin x$



Clicker Question 18-4

Consider now the differential equation $\dot{x} = x^3 - x$. To find $t(x)$, we can integrate:

$$t(x) = \int_{x_0}^x \frac{dx'}{x'^3 - x'}$$

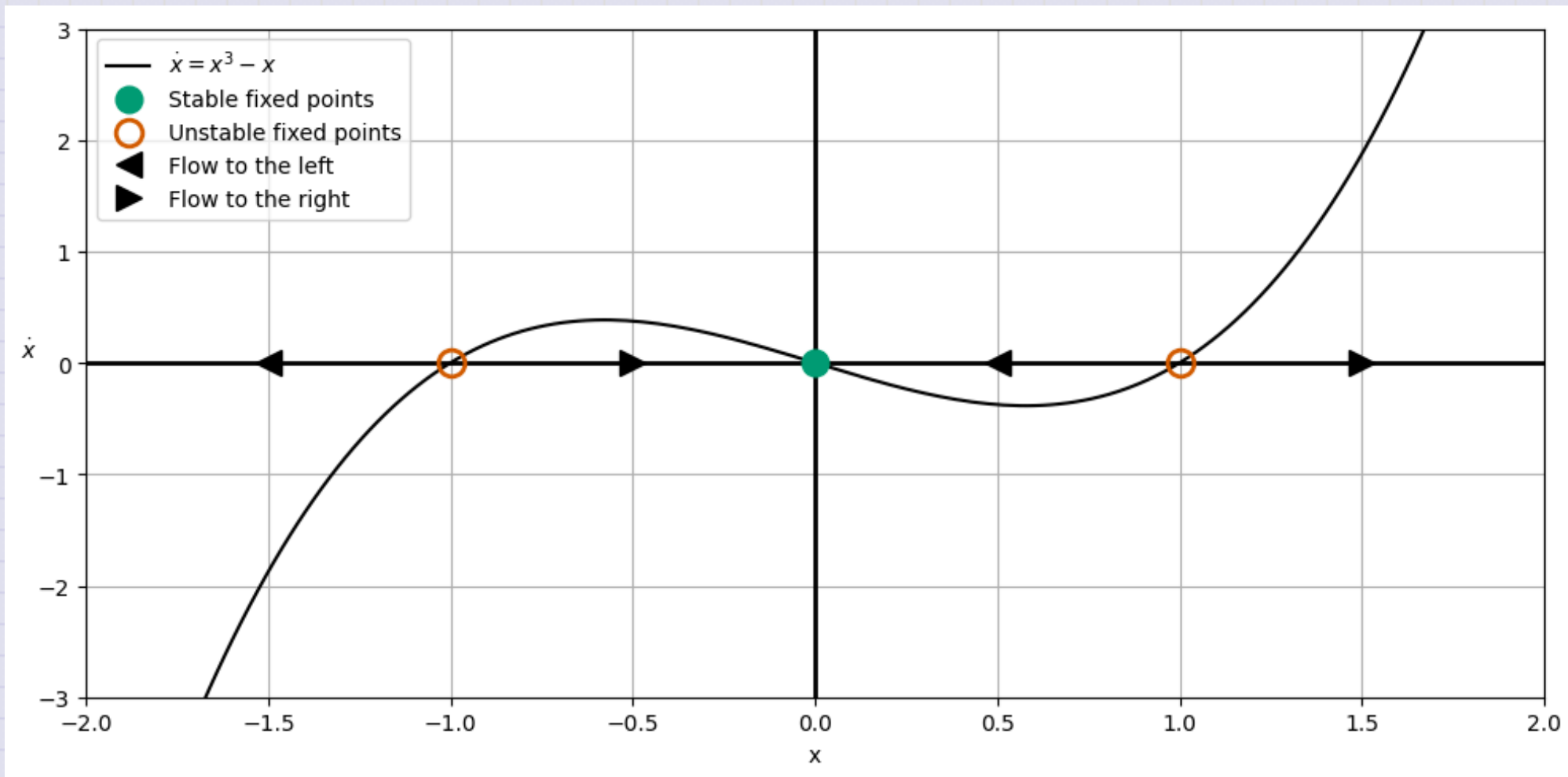
That yields the following solution (😬😬😬):

$$t(x) = \left(\frac{1}{2} \ln(1 - x^2) - \ln(x) \right) - \left(\frac{1}{2} \ln(1 - x_0^2) - \ln(x_0) \right)$$

1. Find the equilibrium points (x^*) of the system.
2. Sketch the differential equation $\dot{x} = x^3 - x$ in the phase space x vs. $\dot{x}$.
3. What can you say about the stability of the critical points? Add these to your plot.

Click when you and your table are done.

Phase Space Diagram for $\dot{x} = x^3 - x$



The Harmonic Oscillator Gets a Bad Rap

The SHO is a linear system. It's boring. It's predictable. It's stable. But it can help us understand nonlinear 2nd order ODEs and thus more complex systems.

Consider the physical pendulum. The equation of motion is

$$mL^2\ddot{\theta} = -mgL \sin \theta$$

Or more simply:

$$\ddot{\theta} = -\frac{g}{L} \sin \theta$$

In the case of small angles, $\sin \theta \approx \theta$, and we have a linear system.

$$\ddot{\theta} = -\frac{g}{L} \theta$$

Examples of SHOs

- The spring-mass system $\ddot{x} = -\omega^2 x$
- The simple pendulum $\ddot{\theta} = -\frac{g}{L} \theta$
- The LC circuit $\ddot{q} = -\frac{1}{LC} q$
- Water in a u-tube $\ddot{h} = -\frac{2g\rho A}{M} h$
- A jump rope $\ddot{u} = \frac{T}{\lambda} \left(\frac{n\pi}{d} \right)^2 u$

Any system with a local minimum in the potential energy can be modeled as an SHO.